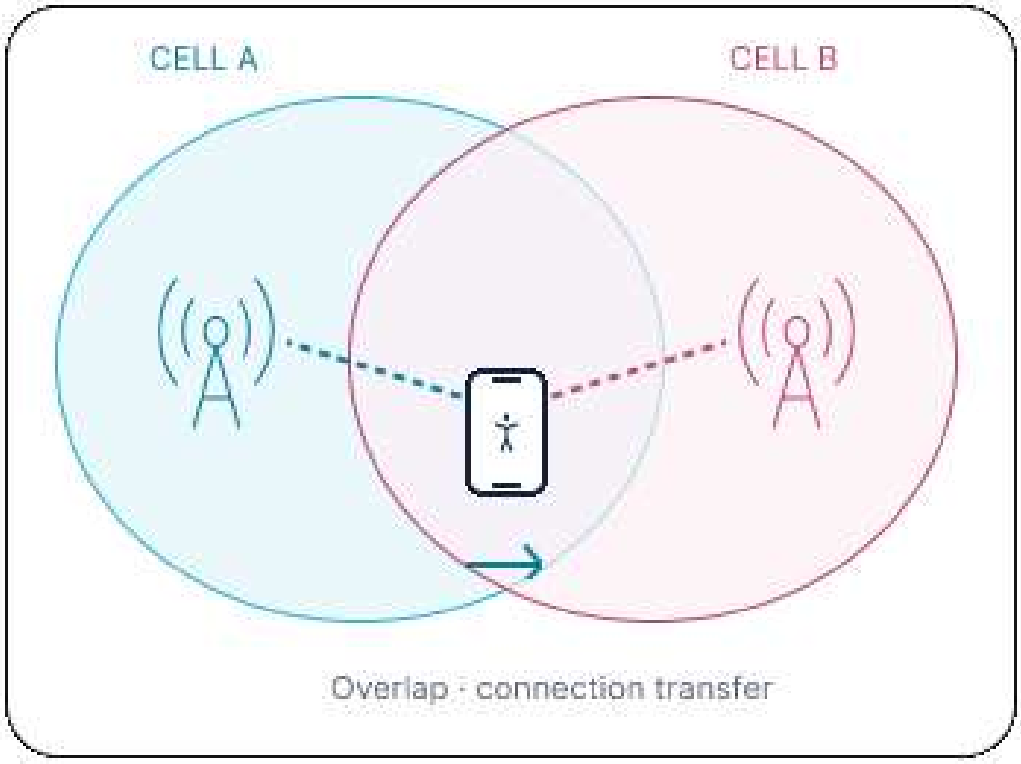
A GUIDE TO SEAMLESS MOBILITY

HANDOFF IN MOBILE NETWORK

Moving between cells. Staying connected.



Overlap · connection transfer

Subject: Mobile Communication

Topic: Handoff in Mobile Networks

▷ Start Simulation

Six-part lesson · Illustrated, not interactive

Handoff

An ongoing voice call or data session moves from one cell or base station to another without interrupting communication as the device moves.

From movement to continuity

FLOW

01 Mobile Device

↓

02 Current Base Station

↓

03 Signal Strength Monitoring

↓

04 Target Base Station Selection

↓

05 Connection Transfer

↓

06 Continuous Communication

Handoff Process

The network monitors signal quality and selects a suitable neighboring station for transfer.

Why It Matters

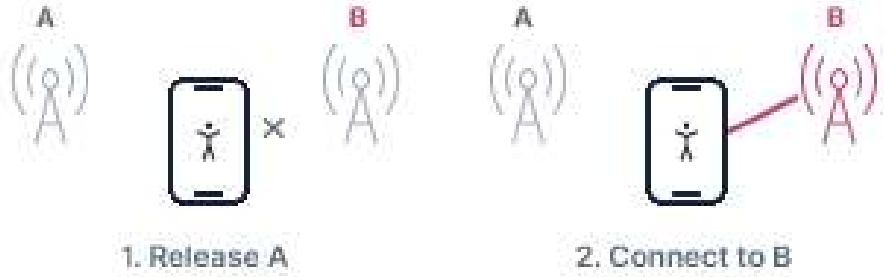
Keeps sessions continuous and supports mobility across cellular coverage areas.

Handoff Types

Two connection strategies. One shared goal: keep the session moving.

Hard handoff

BREAK BEFORE MAKE

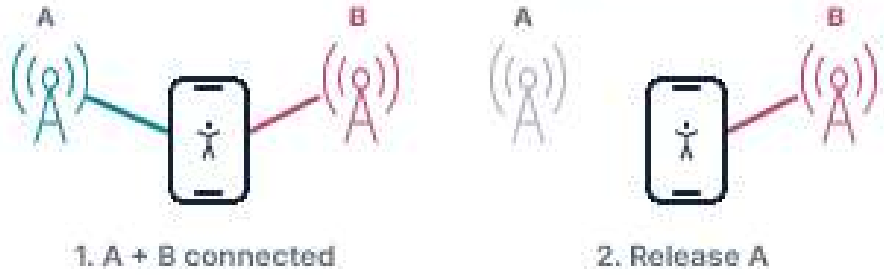


1. Release A 2. Connect to B

The source link ends before the target link is established. A brief interruption can occur.

Soft handoff

MAKE BEFORE BREAK



1. A + B connected 2. Release A

The device connects to the target while keeping the source link, then releases the source.

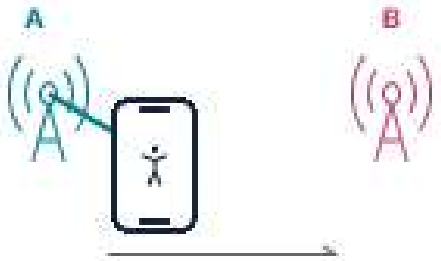
TECHNOLOGY CONTEXT

Soft handoff is associated with CDMA. LTE and 5G generally use hard handover.

Connection Transfer

Follow a device from serving Cell A to target Cell B as it moves through the overlap.

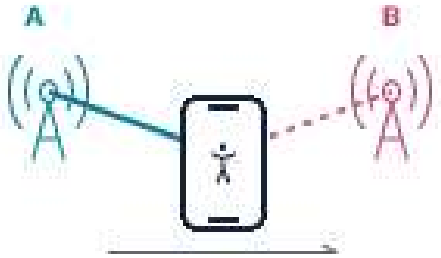
ILLUSTRATED WALKTHROUGH · STATIC



01 Serving Cell A

A carries the session. B is a neighbor.

A strong · B weak



02 Overlap zone

Measure neighbors; prepare target B.

A falling ↘ · B rising ↗



03 Cell B connected

B carries the session. Source A is released.

B strong · A weak

What happens in the network?

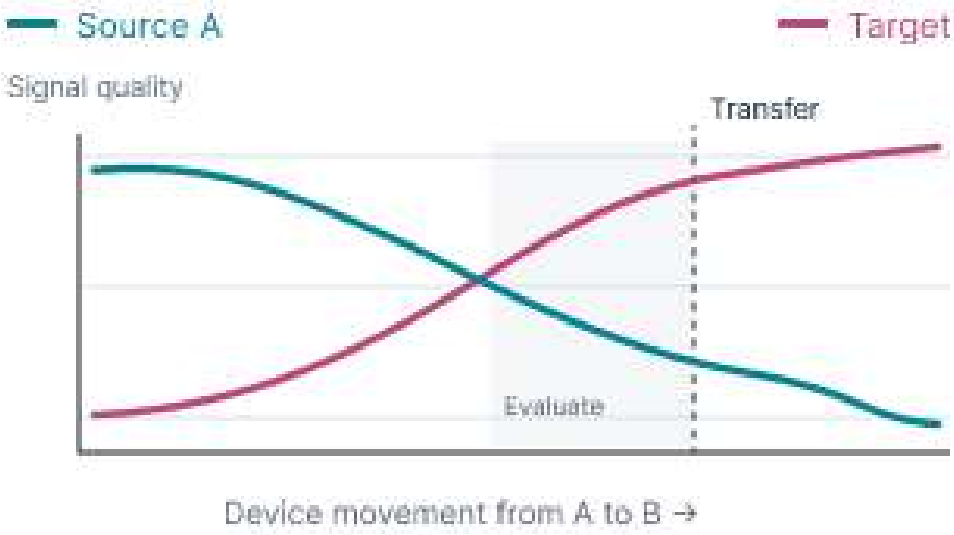
Measurement report → target preparation → handover command → access target → release source

Solid = serving link · Dashed = preparation

Timing and criteria depend on the network.

Handoff Decision

A handoff is a policy-guided decision, not simply a switch to the nearest tower.



Illustrative, not measured data. No universal signal threshold is implied.

01 **Signal quality**
Compare serving and neighbor measurements.

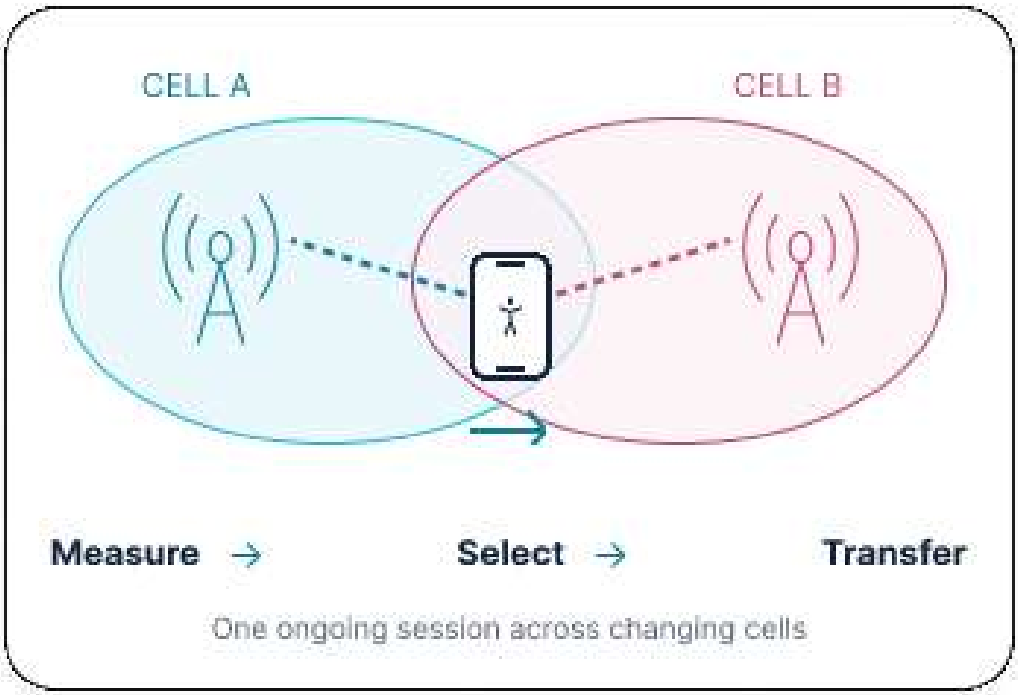
02 **Hysteresis + time-to-trigger**
Require a margin and persistence to limit ping-pong.

03 **Mobility, load + service needs**
Consider speed, capacity and session requirements.

Use trends and context—not distance alone or a single signal reading.

Summary

Seamless mobility is the goal: keep the session going, even when the serving cell changes.



What it enables

Session continuity, wider mobility and a more consistent user experience.

What needs care

Brief interruption risk, ping-pong between cells and handover failure.

THE KEY TAKEAWAY

**The connection changes.
The conversation continues.**